PHYSICAL CUME
SEPTEMBER 24, 2004

1.) Write an expression for the molecular rotational partition function of an unsymmetrical linear molecule, such as HCl, treated as a rigid rotor.

The energy levels of a rigid rotor are $E_J = hcBJ(J+1)$, where J = 0, 1, 2, ... , and B is the rotational constant. (20% credit)

2.) What general qualitative statement can you make about the moments of inertia, the rotational energy levels, the mass, and the size of molecules whose molecular rotational partition functions at room temperature are large? (20% credit)

3.) Evaluate the molecular rotational partition function for $^{1}H^{35}Cl$ at 25°C, with a 1% precision, given that B=10.59 cm^{-1}.

The masses are ^{1}H=1.008 amu (atomic mass unit); ^{35}Cl=35.45 amu; k=1.38066 E-23 JK^{-1}; h=6.62608 E-34 Js; c=2.998 E+8 ms^{-1}. (60% credit)